



E COMMERCE CUSTOMER BEHAVIOUR ANALYSIS

Power Bi Project

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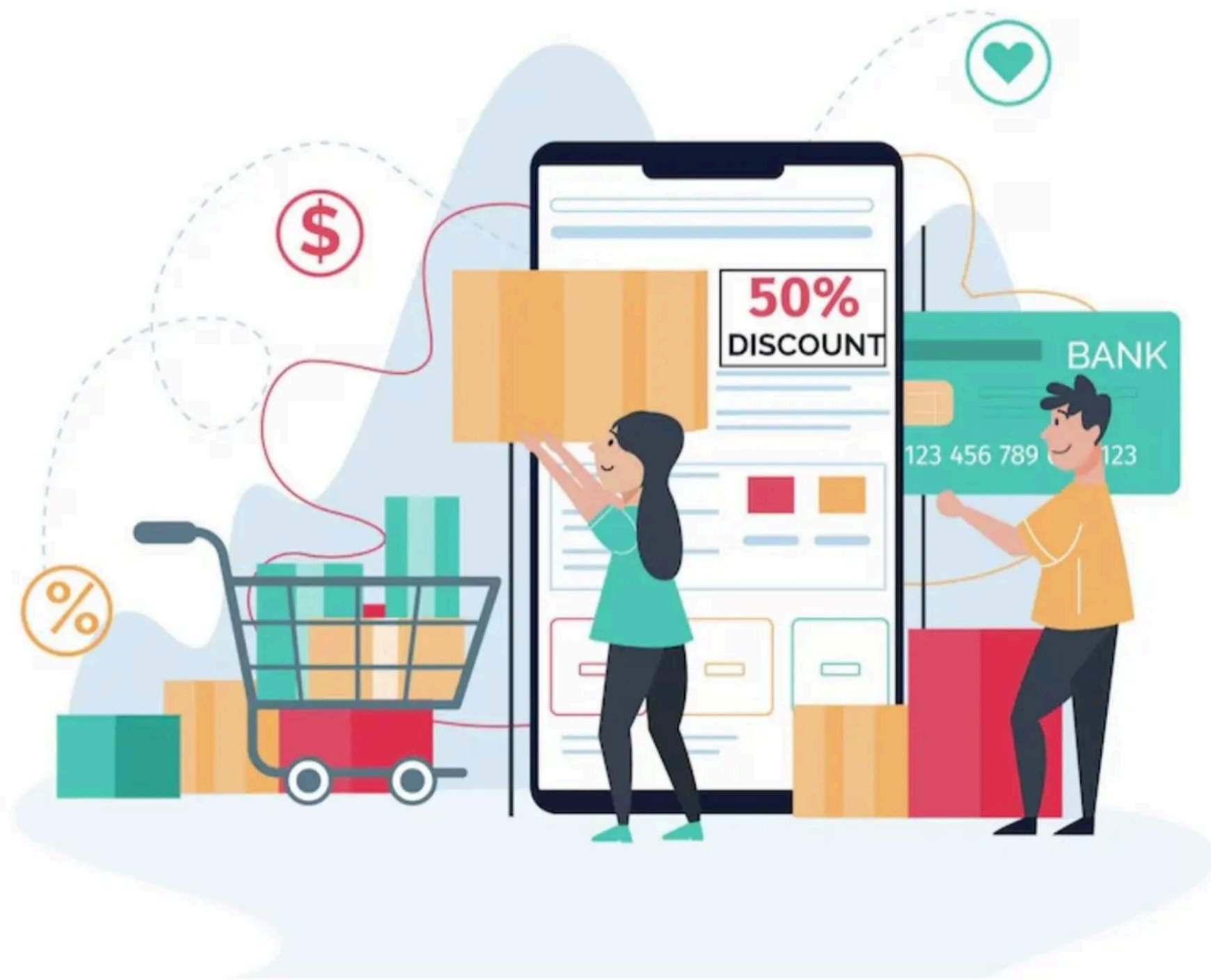
CONTENTS

- INTRODUCTION
- ANALYSIS
- DASHBOARD
- KEY INSIGHTS
- CONCLUSION



INTRODUCTION

- This project focuses on analyzing customer purchasing behavior in an e-commerce environment.
- The dashboard provides insights into sales performance, customer demographics, payment preferences, and purchasing patterns.
- Interactive visualizations and KPIs help transform raw transaction data into meaningful business insights.
- The dashboard supports understanding customer preferences and buying trends.
- The primary objective is to assist businesses in improving customer engagement and sales performance through data-driven decision-making.



ANALYSIS (KPIs AND CHARTS)

1. Total Sales – 233K

- The business generated total sales of 233K.
- This represents the overall revenue generated from customer purchases.
- Sales performance reflects customer demand across different product categories.

2. Total Customers – 4K

- The dataset contains approximately 4,000 customers.
- This provides a broad customer base for behavioral analysis.
- Understanding customer preferences helps improve marketing strategies.

3. Average Rating – 3.75

- Customers provided an average rating of 3.75.
- This indicates moderate customer satisfaction.
- Product quality and customer service play an important role in maintaining ratings.

4. Average Purchase – 59.8

- The average purchase amount is 59.8 USD.
- This reflects the typical spending behavior of customers.
- Increasing average purchase value can improve overall revenue.

Total Sale by Category



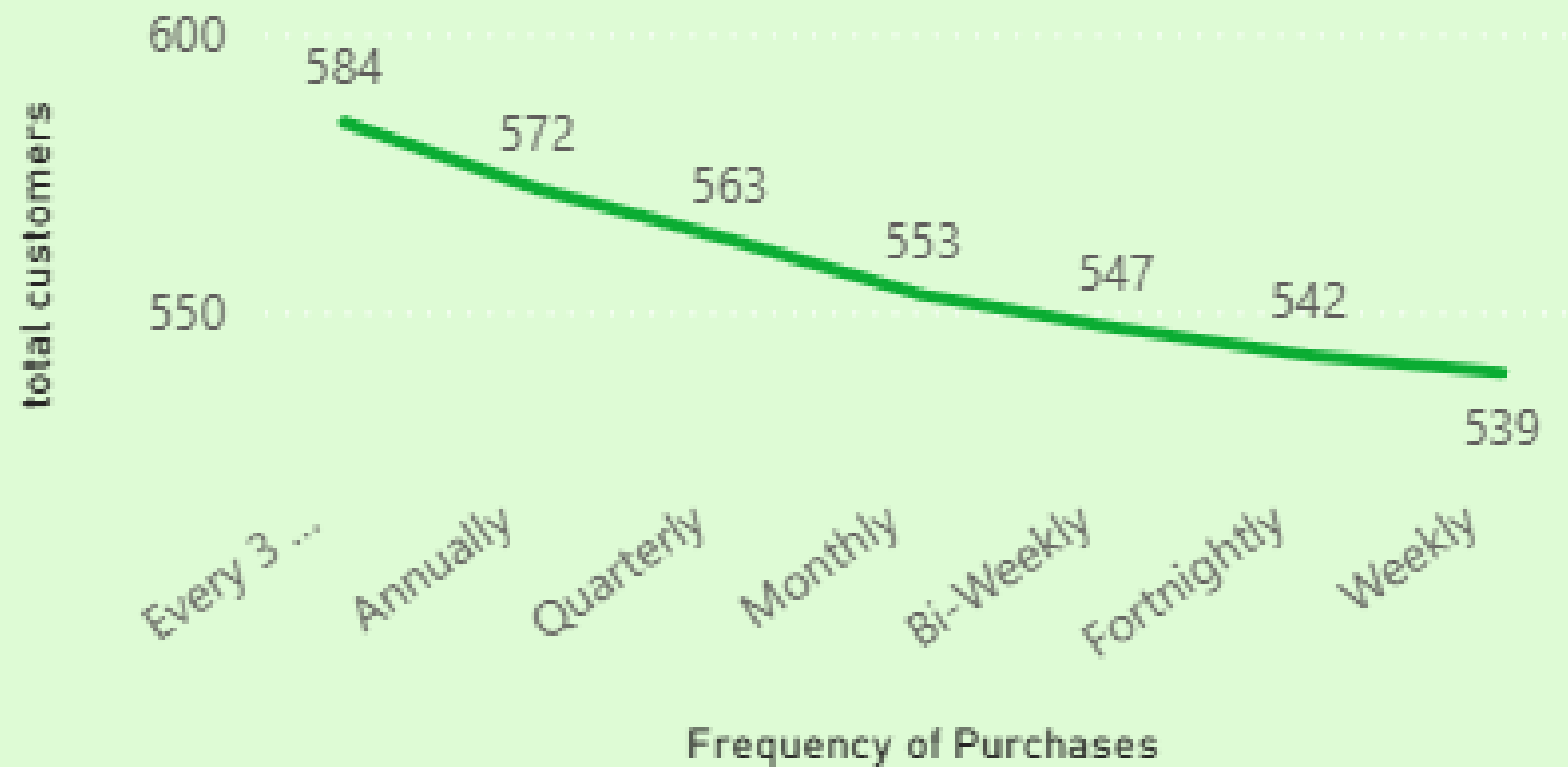
Insight

- Clothing contributes the highest sales among all product categories.
- Accessories generate the second-highest revenue.
- Footwear and Outerwear account for comparatively lower sales.

Conclusion

- Clothing remains the most preferred product category among customers.
- Businesses should focus on expanding high-performing categories.
- Marketing efforts can help improve sales in lower-performing segments.

Total customers by Frequency of Purchases



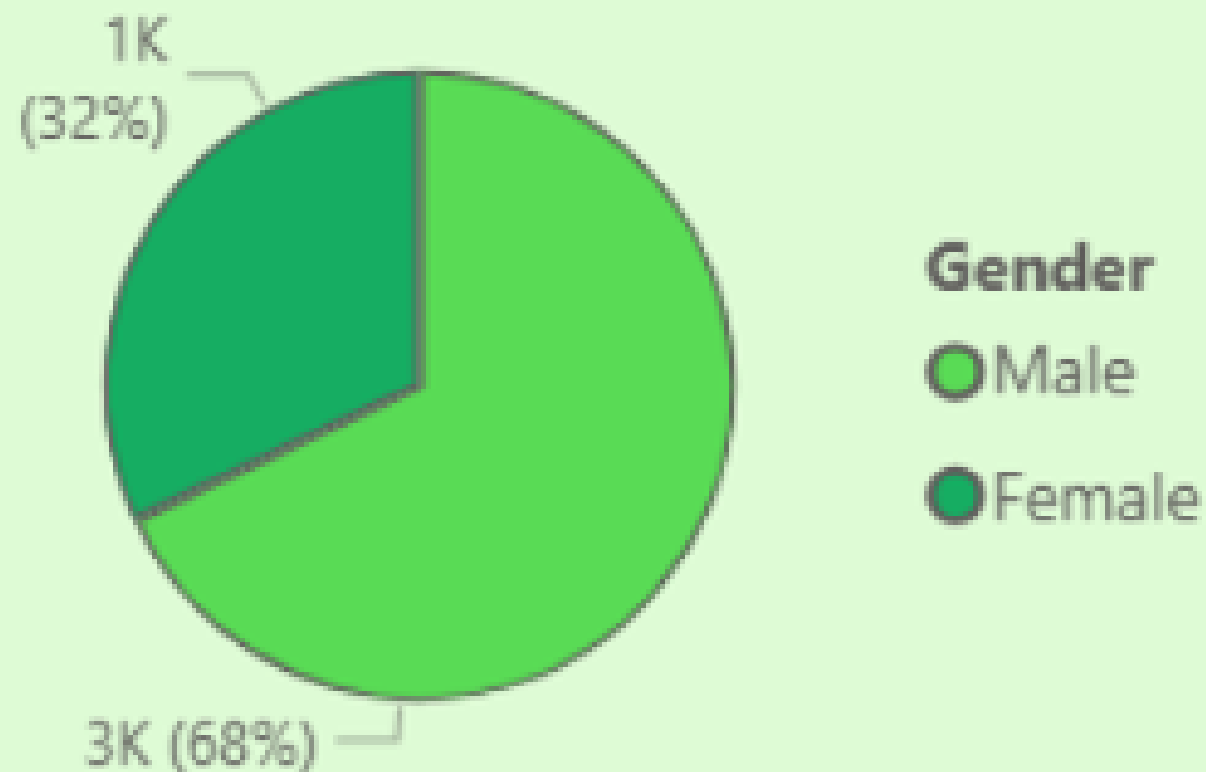
Insight

- Customer count gradually decreases as purchase frequency increases.
- Occasional buyers represent a larger portion of the customer base.
- Frequent buyers form a smaller but valuable customer segment.

Conclusion

- Customer retention strategies are essential for increasing repeat purchases.
- Loyalty programs can encourage customers to buy more frequently.
- Repeat customers contribute significantly to long-term business growth.

Total customers by Gender



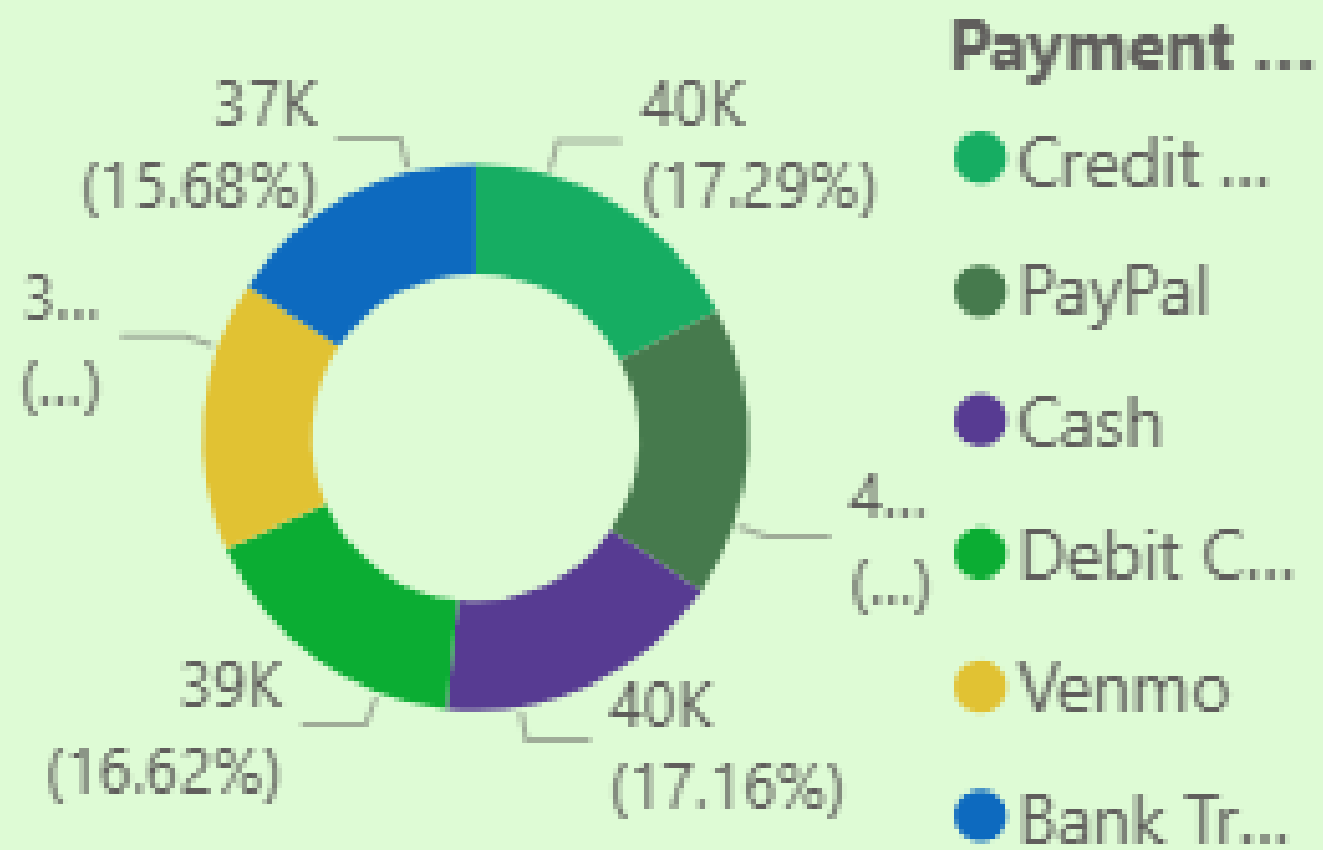
Insight

- Male customers represent the majority of the customer base.
- Female customers account for a smaller proportion of purchases.
- Gender-based purchasing behavior shows clear differences.

Conclusion

- Male customers are a key driver of sales performance.
- Personalized marketing campaigns can improve customer engagement.
- Understanding demographic preferences supports targeted promotions.

Total Sale by Payment Method



Insight

- Sales are distributed relatively evenly across payment methods.
- Credit Cards, Debit Cards, PayPal, Cash, Venmo, and Bank Transfers are commonly used.
- Customers prefer having multiple payment options available.

Conclusion

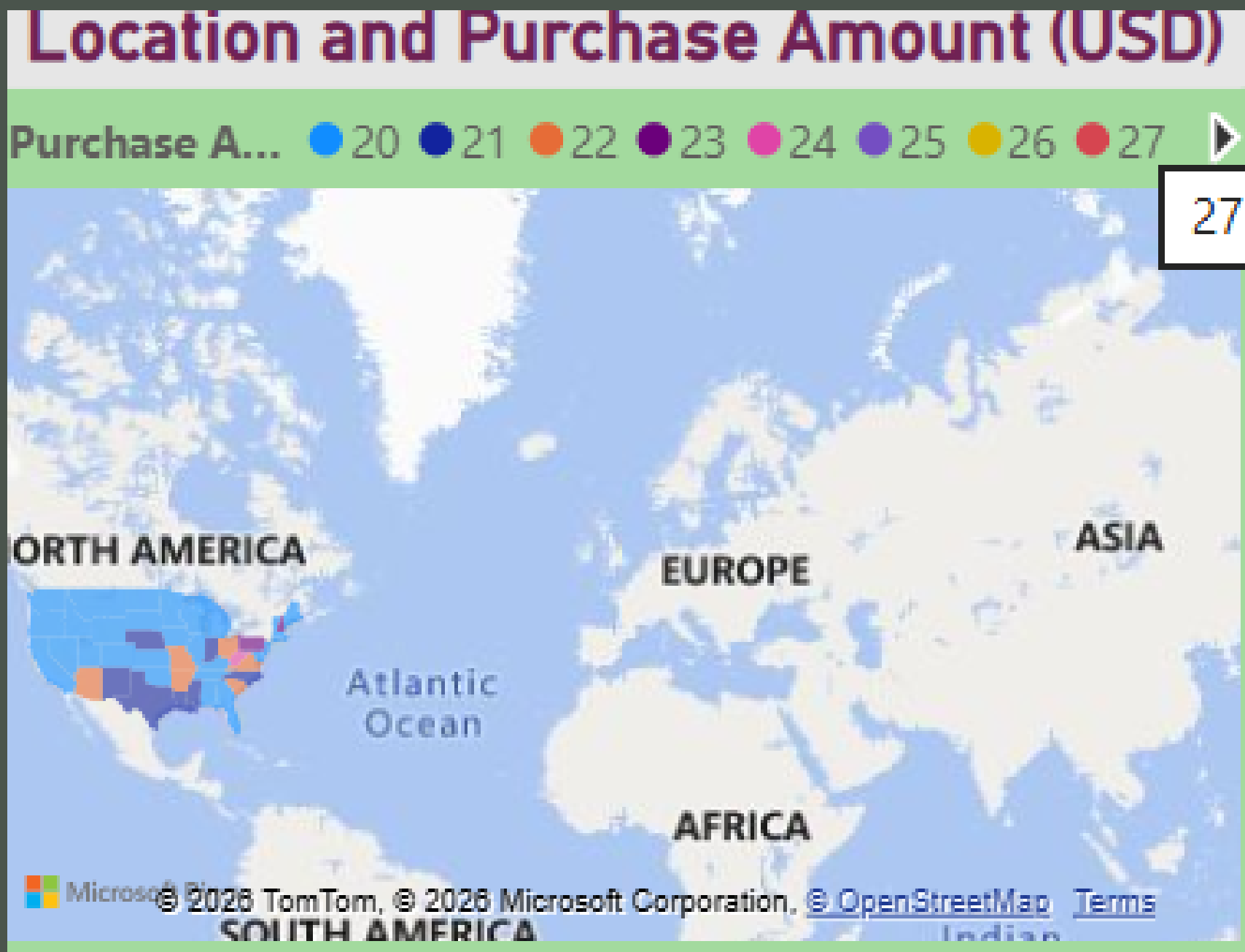
- Offering diverse payment methods enhances customer convenience.
- Flexible payment options can improve conversion rates.
- Businesses should continue supporting multiple secure payment channels.

Insight

- Purchase amounts vary across different geographic locations.
- Certain regions contribute significantly higher revenue.
- Customer spending behavior differs by location.

Conclusion

- Geographic analysis helps identify high-performing markets.
- Regional marketing strategies can improve sales performance.
- Businesses can focus investments on areas with strong customer demand.



Top 7 Customers based on Total purchase

Count of Customer ID	Category	Sum of Review Rating	Frequency of Purchases	Total purchase
5	Clothing	16.40	Every 3 Months	500
4	Clothing	15.80	Fortnightly	400
3	Clothing	9.90	Annually	300
3	Clothing	11.50	Monthly	300
3	Outerwear	13.40	Quarterly	300
2	Accessories	8.50	Every 3 Months	200
2	Accessories	8.90	Weekly	200
2	Footwear	5.50	Annually	200
2	Outerwear	8.40	Annually	200
1	Accessories	3.00	Annually	100
1	Accessories	3.20	Fortnightly	100
1	Accessories	2.80	Monthly	100
1	Clothing	3.50	Quarterly	100
1	Footwear	4.00	Bi-Weekly	100
1	Footwear	3.80	Monthly	100
1	Footwear	4.40	Weekly	100
1	Outerwear	2.70	Every 3 Months	100
1	Outerwear	3.10	Fortnightly	100
1	Outerwear	4.00	Weekly	100
36		132.80		3600

Insight

- A small group of customers contributes a significant portion of total revenue.
- High-value customers demonstrate strong purchasing frequency.
- Customer loyalty influences overall sales performance.

Conclusion

- Retaining top customers is essential for sustainable growth.
- Reward programs and personalized offers can strengthen customer loyalty.
- Businesses should prioritize high-value customer relationships.

DASHBOARD OVERVIEW

E COMMERCE CUSTOMER BEHAVIOUR DASHBOARD



Total Sales

233K

Average Rating

3.75

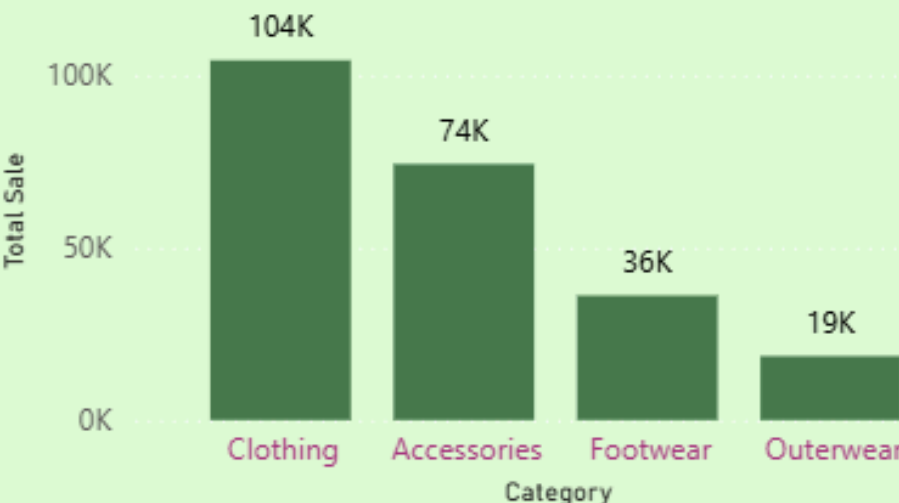
Total customers

4K

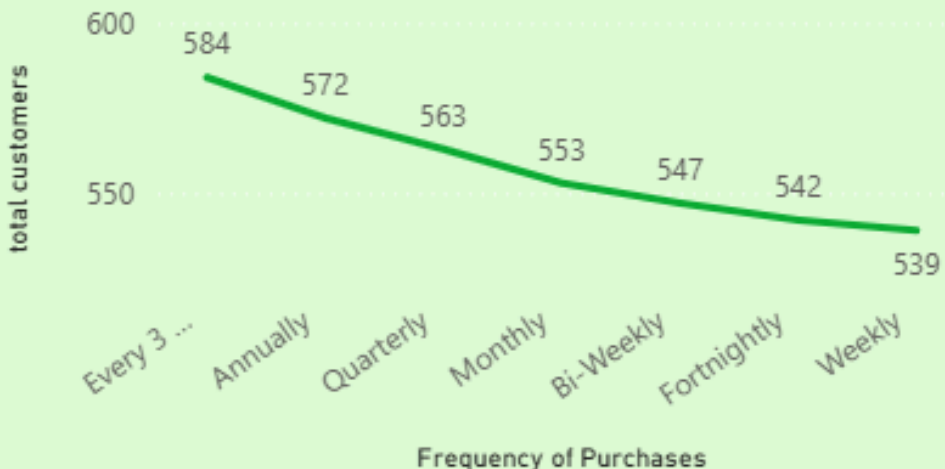
Average Purchase

59.8

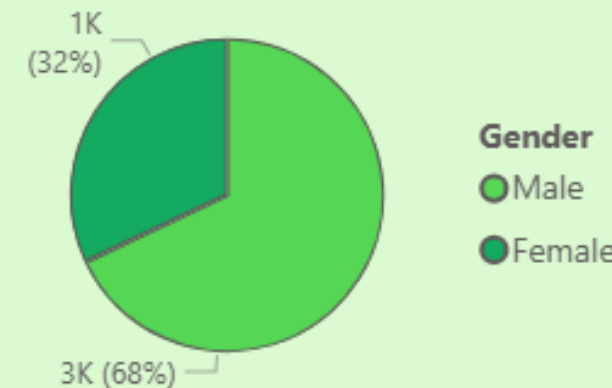
Total Sale by Category



Total customers by Frequency of Purchases



Total customers by Gender



Subscription Status

All

Size

All

Frequency of Purcha...

All

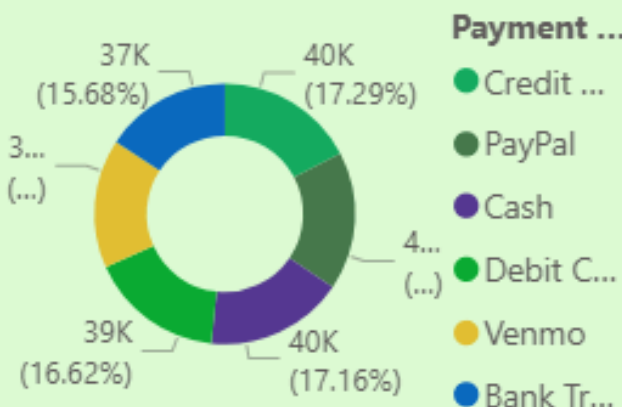
Color

All

Discount Applied

All

Total Sale by Payment Method



Location and Purchase Amount (USD)



Top 7 Customers based on Total purchase

Count of Customer ID	Category	Sum of Review Rating	Frequency
5	Clothing	16.40	Every 3 months
4	Clothing	15.80	Fortnightly
3	Clothing	9.90	Annually
3	Clothing	11.50	Monthly
3	Outerwear	13.40	Quarterly
2	Accessories	8.50	Every 3 months
2	Accessories	8.90	Weekly
2	Footwear	5.50	Annually
2	Outerwear	8.40	Annually
36		132.80	

KEY INSIGHTS

- Total sales reached 233K.
- The business serves approximately 4,000 customers.
- Average customer rating is 3.75.
- Average purchase amount is 59.8 USD.
- Clothing is the highest-selling product category.
- Male customers account for the majority of purchases.
- Payment methods are evenly distributed among customers.
- Purchase behavior varies significantly by location.
- Repeat customers contribute substantial revenue.
- Customer retention remains critical for long-term growth.

CONCLUSION

- The E-Commerce Customer Behaviour Dashboard provides valuable insights into customer preferences, purchasing habits, and sales performance. The analysis highlights that Clothing is the leading product category, Male customers represent a major customer segment, and customer purchase frequency directly impacts business revenue.
- The dashboard also demonstrates the importance of offering multiple payment methods, understanding regional buying patterns, and maintaining strong customer relationships. By leveraging these insights, businesses can improve customer satisfaction, increase sales, and develop more effective marketing strategies.
- Continuous monitoring of customer behavior enables organizations to make informed decisions, enhance customer experiences, and achieve sustainable business growth.

THANK YOU

